

# Fraction Universe

Every graded set, every week — 6 weeks, 30 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

WEEK 6

5 sets

## HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

# One Whole, Cut Up

Answers are whole numbers — when a question asks for a numerator, type just that number.

## Warm-Up 12 ITEMS • RECALL

1 Halves in one whole

\_\_\_\_\_

2 Quarters in one whole

\_\_\_\_\_

3 Eighths in one whole

\_\_\_\_\_

4  $1/2 = ?/4 \rightarrow$  numerator

\_\_\_\_\_

5  $1/2 = ?/8 \rightarrow$  numerator

\_\_\_\_\_

6  $3/3 =$  how many wholes?

\_\_\_\_\_

7 Thirds in one whole

\_\_\_\_\_

8 Fifths in one whole

\_\_\_\_\_

9 Sixths in one whole

\_\_\_\_\_

10  $1/3 = ?/6 -$  numerator

\_\_\_\_\_

11  $1/5 = ?/10 -$  numerator

\_\_\_\_\_

12 How many thirds in 2 wholes?

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $5/5 =$  how many wholes?(Count the quarters)

.....

14.  $7/4 -$  how many whole ones?(Three in each whole)

.....

15.  $7/4 -$  how many quarters left over?(All five pieces)

.....

16.  $3/4 = ?/12 -$  numerator(4/4 makes one whole)

.....

17.  $2/5 = ?/20 -$  numerator(7 - 4)

.....

18.  $5/6 = ?/12 -$  numerator

.....

19.  $7/8 = ?/16 -$  numerator

.....

20.  $2/3 = ?/15 -$  numerator

.....

21.  $1/4 + 1/4 + 1/4 = ?/4 \rightarrow$  numerator

.....

22. Marks between 0 and 1 when you cut into fifths (not counting 0 and 1)

.....

**★ Challenge** NOT GRADED • REASON IT OUT

23.  $\frac{3}{8}$  and  $\frac{5}{8}$  — how many eighths apart? (Five pieces need four cuts)

24. Tenths in one whole (5 - 3)

25. Cuts needed to make eighths

26.  $\frac{2}{9}$  and  $\frac{7}{9}$  — how many ninths apart

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Same Number, New Name

Multiply top and bottom by the same thing and nothing about the amount changes.

## Warm-Up 12 ITEMS • RECALL

1  $1/2 = ?/6 \rightarrow$  numerator

\_\_\_\_\_

2  $1/3 = ?/6 \rightarrow$  numerator

\_\_\_\_\_

3  $2/3 = ?/6 \rightarrow$  numerator

\_\_\_\_\_

4  $1/4 = ?/8 \rightarrow$  numerator

\_\_\_\_\_

5  $3/4 = ?/8 \rightarrow$  numerator

\_\_\_\_\_

6  $1/5 = ?/10 \rightarrow$  numerator

\_\_\_\_\_

7  $1/2 = ?/10 \rightarrow$  numerator

\_\_\_\_\_

8  $1/4 = ?/12 \rightarrow$  numerator

\_\_\_\_\_

9  $1/3 = ?/9 \rightarrow$  numerator

\_\_\_\_\_

10  $2/3 = ?/9 \rightarrow$  numerator

\_\_\_\_\_

11  $3/5 = ?/10 \rightarrow$  numerator

\_\_\_\_\_

12  $1/6 = ?/12 \rightarrow$  numerator

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $3/4 = ?/12 \rightarrow$  numerator (The bottom was tripled)

\_\_\_\_\_

14.  $2/5 = ?/20 \rightarrow$  numerator (The bottom was multiplied by 4)

\_\_\_\_\_

15.  $6/8$  in quarters =  $?/4 \rightarrow$  numerator (Divide both by 2)

\_\_\_\_\_

16.  $10/15$  simplified  $\rightarrow$  numerator (Divide both by 5)

\_\_\_\_\_

17.  $9/12$  simplified  $\rightarrow$  denominator (Divide both by 3)

\_\_\_\_\_

18.  $6/8 = ?/4 \rightarrow$  numerator

\_\_\_\_\_

19.  $10/15 = ?/3 \rightarrow$  numerator

\_\_\_\_\_

20.  $12/16 = ?/4 \rightarrow$  numerator

\_\_\_\_\_

21.  $9/15 = ?/5 \rightarrow$  numerator

\_\_\_\_\_

22.  $14/21 = ?/3 \rightarrow$  numerator

\_\_\_\_\_



# Which Is Bigger

Make the pieces the same size first. Then it's just counting.

## Warm-Up 12 ITEMS • RECALL

1 Bigger:  $\frac{1}{2}$  or  $\frac{1}{4}$  → its denominator  
\_\_\_\_\_

2 Bigger:  $\frac{1}{3}$  or  $\frac{1}{5}$  → its denominator  
\_\_\_\_\_

3 Bigger:  $\frac{3}{4}$  or  $\frac{1}{4}$  → its numerator  
\_\_\_\_\_

4 Bigger:  $\frac{2}{5}$  or  $\frac{3}{5}$  → its numerator  
\_\_\_\_\_

5 Bigger:  $\frac{5}{6}$  or  $\frac{5}{8}$  → its denominator  
\_\_\_\_\_

6 Bigger:  $\frac{7}{10}$  or  $\frac{3}{10}$  → its numerator  
\_\_\_\_\_

7 Bigger:  $\frac{1}{2}$  or  $\frac{1}{5}$  — its denominator  
\_\_\_\_\_

8 Bigger:  $\frac{1}{4}$  or  $\frac{1}{6}$  — its denominator  
\_\_\_\_\_

9 Bigger:  $\frac{4}{5}$  or  $\frac{2}{5}$  — its numerator  
\_\_\_\_\_

10 Bigger:  $\frac{5}{8}$  or  $\frac{3}{8}$  — its numerator  
\_\_\_\_\_

11 Bigger:  $\frac{1}{3}$  or  $\frac{1}{10}$  — its denominator  
\_\_\_\_\_

12 Bigger:  $\frac{7}{10}$  or  $\frac{3}{10}$  — its numerator  
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $\frac{2}{3}$  or  $\frac{3}{5}$  in fifteenths → bigger numerator (10/15 against 9/15) .....

14.  $\frac{3}{4}$  or  $\frac{5}{8}$  in eighths → bigger numerator (6/8 against 5/8) .....

15.  $\frac{5}{6}$  or  $\frac{7}{9}$  in eighteenths → bigger numerator (15/18 against 14/18) .....

16. Closer to 1:  $\frac{7}{8}$  or  $\frac{5}{6}$  → its denominator (A gap of  $\frac{1}{8}$  beats a gap of  $\frac{1}{6}$ ) .....

17.  $\frac{2}{3}$  and  $\frac{3}{4}$  in twelfths → bigger numerator (8/12 against 9/12) .....

18. Bigger:  $\frac{2}{3}$  or  $\frac{3}{5}$  in fifteenths — its numerator .....

19. Bigger:  $\frac{3}{4}$  or  $\frac{5}{8}$  in eighths — its numerator .....

20. Bigger:  $\frac{5}{6}$  or  $\frac{4}{5}$  in thirtieths — its numerator .....

21. Bigger:  $\frac{1}{2}$  or  $\frac{4}{9}$  in eighteenths — its numerator .....

22. Bigger:  $\frac{7}{12}$  or  $\frac{1}{2}$  in twelfths — its numerator .....



26. Halfway between  $\frac{1}{4}$  and  $\frac{1}{2}$  as  $\frac{?}{8}$  — numerator



# Adding Needs Matching Pieces

Change one fraction, or both, until the bottoms agree. Then count.

## Warm-Up 12 ITEMS • RECALL

1  $7/10 - 4/10 = ?/10 \rightarrow$   
numerator

\_\_\_\_\_

2  $7/10 - 3/10 = ?/10 -$   
numerator

\_\_\_\_\_

3 A whole cut into 2 equal  
parts — one part is 1 over  
what

\_\_\_\_\_

4 A whole cut into 3 equal  
parts — one part is 1 over  
what

\_\_\_\_\_

5 A whole cut into 4 equal  
parts — one part is 1 over  
what

\_\_\_\_\_

6 A whole cut into 6 equal  
parts — one part is 1 over  
what

\_\_\_\_\_

7 A whole cut into 8 equal  
parts — one part is 1 over  
what

\_\_\_\_\_

8 How many 2ths make one  
whole

\_\_\_\_\_

9 How many 3ths make one  
whole

\_\_\_\_\_

10 How many 4ths make  
one whole

\_\_\_\_\_

11 How many 6ths make  
one whole

\_\_\_\_\_

12 How many 8ths make  
one whole

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Which is the bigger unit fraction:  $1/2$  or  $1/4$  — give the denominator ( $2/4 + 1/4$ )

\_\_\_\_\_

14. Which is the bigger unit fraction:  $1/3$  or  $1/6$  — give the denominator ( $4/6 + 1/6$ )

\_\_\_\_\_

15. Which is the bigger unit fraction:  $1/4$  or  $1/8$  — give the denominator ( $6/8 - 1/8$ )

\_\_\_\_\_

16. Which is the bigger unit fraction:  $1/2$  or  $1/6$  — give the denominator ( $5/6 - 2/6$ )

\_\_\_\_\_

17. Which is the bigger unit fraction:  $1/3$  or  $1/4$  — give the denominator ( $3/6 + 2/6$ )

\_\_\_\_\_

18.  $3/4$  — how many more 4ths to reach one whole

\_\_\_\_\_

19.  $5/6$  — how many more 6ths to reach one whole

\_\_\_\_\_

20.  $7/8$  — how many more 8ths to reach one whole

\_\_\_\_\_

21.  $2/3$  — how many more 3ths to reach one whole

\_\_\_\_\_

22.  $1/2$  of 12

\_\_\_\_\_





## Working space



## Error journal

# Fraction of a Set & Closest to One

Enrichment day. Divide by the bottom, multiply by the top — always that order.

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 **Challenge** NOT GRADED • REASON IT OUT

1. How many halves make one whole ( $20 \div 4$ )
2. How many thirds make one whole ( $30 \div 5 = 6$ , then  $\times 2$ )
3.  $4/4$  — how many wholes ( $40 \div 8 = 5$ , then  $\times 3$ )
4. Bigger unit fraction:  $1/2$  or  $1/4$  — the denominator ( $45 \div 3 = 15$ , then  $\times 2$ )
5. Bigger:  $1/2$  or  $1/3$  — type it as a/b ( $42 \div 6 = 7$ , then  $\times 5$ )
6. Bigger:  $1/4$  or  $1/8$  — type it as a/b ( $18 \div 3 = 6$  is one quarter)
7. Bigger:  $2/3$  or  $3/4$  — type it as a/b (Gaps of  $1/6$ ,  $1/8$ ,  $1/10$ )
8. Bigger:  $3/8$  or  $1/2$  — type it as a/b ( $16 \div 3$ )
9. Bigger:  $5/6$  or  $2/3$  — type it as a/b
10.  $6/6$  — how many wholes
11.  $8/8$  — how many wholes
12.  $3/3$  — how many wholes
13.  $5/4$  — how many whole ones are in it
14.  $7/6$  — how many whole ones are in it

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

# Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 1.1 • One Whole, Cut Up

OUT OF 20

Warm-Up • 1-11 • 2 | 4 | 8 | 2 | 4 | 1 | 3 | 5 | 6 | 2 | 2

Core • 12-20 • 6 | 1 | 1 | 3 | 9 | 8 | 10 | 14 | 10

Challenge • 21-26 • 3 | 4 | 2 | 10 | 7 | 5

## 1.2 • Same Number, New Name

OUT OF 22

Warm-Up • 1-12 • 3 | 2 | 4 | 2 | 6 | 2 | 5 | 3 | 3 | 6 | 6 | 2

Core • 13-22 • 9 | 8 | 3 | 2 | 4 | 3 | 2 | 3 | 3 | 2

Challenge • 23-26 • 2 | 9 | 6 | 5

## 1.3 • Which Is Bigger

OUT OF 22

Warm-Up • 1-12 • 2 | 3 | 3 | 3 | 6 | 7 | 2 | 4 | 4 | 5 | 3 | 7

Core • 13-22 • 10 | 6 | 15 | 8 | 9 | 10 | 6 | 25 | 9 | 7

Challenge • 23-26 • 21 | 5 | 16 | 3

## 1.4 • Adding Needs Matching Pieces

OUT OF 24

Warm-Up • 1-2 • 3 | 4

Core • 3-24 • 2 | 3 | 4 | 6 | 8 | 2 | 3 | 4 | 6 | 8 | 2 | 3 | 4 | 2 | 3 | 1 | 1 | 1 | 1 | 6 | 3 | 4

Challenge • 25-48 • 2 | 3 | 5 | 3 | 1 | 3 | 5 | 5 | 3 | 5 | 17 | 7 | 2 | 4 | 6 | 3 | 5 | 4 | 7 | 1 | 3 | 7 | 19 | 13

## Fri • Fraction of a Set & Closest to One

ENRICHMENT

Challenge • 1-14 • 5 | 12 | 15 | 30 | 35 | 24 | 10 | 19 | 10 | 8 | 14 | 30 | 18 | 24

# Same Number, New Clothes

Multiply top and bottom by the same thing and the value never moves. Type fractions like  $\frac{3}{4}$ .

## Warm-Up 12 ITEMS • RECALL

1  $\frac{1}{2} = \frac{?}{4}$  — the numerator  
\_\_\_\_\_

2  $\frac{1}{2} = \frac{?}{6}$   
\_\_\_\_\_

3  $\frac{1}{3} = \frac{?}{6}$   
\_\_\_\_\_

4  $\frac{1}{4} = \frac{?}{8}$   
\_\_\_\_\_

5  $\frac{2}{3} = \frac{?}{6}$   
\_\_\_\_\_

6  $\frac{3}{4} = \frac{?}{8}$   
\_\_\_\_\_

7  $\frac{1}{2} = \frac{?}{8}$   
\_\_\_\_\_

8  $\frac{1}{3} = \frac{?}{12}$   
\_\_\_\_\_

9  $\frac{1}{5} = \frac{?}{15}$   
\_\_\_\_\_

10  $\frac{2}{3} = \frac{?}{12}$   
\_\_\_\_\_

11  $\frac{3}{5} = \frac{?}{15}$   
\_\_\_\_\_

12  $\frac{1}{4} = \frac{?}{16}$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $\frac{2}{5} = \frac{?}{15}$  .....

14.  $\frac{3}{8} = \frac{?}{24}$  .....

15.  $\frac{5}{6} = \frac{?}{18}$  .....

16.  $\frac{4}{7} = \frac{?}{21}$  .....

17.  $\frac{7}{10} = \frac{?}{100}$  .....

18.  $\frac{4}{5} = \frac{?}{25}$  .....

19.  $\frac{5}{8} = \frac{?}{24}$  .....

20.  $\frac{3}{7} = \frac{?}{21}$  .....

21.  $\frac{5}{9} = \frac{?}{27}$  .....



## Working space



## Error journal

# Simplest Form

Divide top and bottom by the same thing until nothing else fits.

## Warm-Up 11 ITEMS • RECALL

1  $\frac{2}{4}$  in simplest form —  
type as a/b

\_\_\_\_\_

2  $\frac{3}{6}$

\_\_\_\_\_

3  $\frac{4}{8}$

\_\_\_\_\_

4  $\frac{2}{6}$

\_\_\_\_\_

5  $\frac{5}{10}$

\_\_\_\_\_

6  $\frac{3}{9}$

\_\_\_\_\_

7  $\frac{6}{12}$  in simplest form —  
type as a/b

\_\_\_\_\_

8  $\frac{4}{6}$

\_\_\_\_\_

9  $\frac{5}{15}$

\_\_\_\_\_

10  $\frac{8}{10}$

\_\_\_\_\_

11  $\frac{9}{12}$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12.  $\frac{6}{8}$

.....

13.  $\frac{10}{15}$

.....

14.  $\frac{12}{16}$

.....

15.  $\frac{18}{24}$

.....

16.  $\frac{20}{25}$

.....

17.  $\frac{21}{28}$

.....

18.  $\frac{30}{45}$

.....

19.  $\frac{14}{35}$

.....

20.  $\frac{27}{36}$

.....





# Spot the Equivalent

Cross-multiply, or find a common denominator. Either proof counts.

## Warm-Up 12 ITEMS • RECALL

1 Is  $\frac{1}{2}$  the same as  $\frac{2}{4}$  — yes or no

\_\_\_\_\_

2 Is  $\frac{1}{3}$  the same as  $\frac{2}{6}$

\_\_\_\_\_

3 Is  $\frac{1}{2}$  the same as  $\frac{1}{3}$

\_\_\_\_\_

4 Is  $\frac{3}{4}$  the same as  $\frac{6}{8}$

\_\_\_\_\_

5 Is  $\frac{2}{5}$  the same as  $\frac{4}{10}$

\_\_\_\_\_

6 Is  $\frac{2}{3}$  the same as  $\frac{3}{4}$

\_\_\_\_\_

7 Is  $\frac{1}{4}$  the same as  $\frac{2}{8}$  — yes or no

\_\_\_\_\_

8 Is  $\frac{2}{3}$  the same as  $\frac{4}{6}$

\_\_\_\_\_

9 Is  $\frac{1}{2}$  the same as  $\frac{2}{5}$

\_\_\_\_\_

10 Is  $\frac{3}{5}$  the same as  $\frac{6}{10}$

\_\_\_\_\_

11 Is  $\frac{1}{3}$  the same as  $\frac{3}{9}$

\_\_\_\_\_

12 Is  $\frac{2}{4}$  the same as  $\frac{3}{5}$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is  $\frac{6}{9}$  the same as  $\frac{8}{12}$  — yes or no

.....

14. Is  $\frac{5}{8}$  the same as  $\frac{10}{18}$

.....

15. Is  $\frac{9}{12}$  the same as  $\frac{15}{20}$

.....

16.  $\frac{1}{2} \times \frac{5}{5}$  — type as a/b

.....

17. Is  $\frac{7}{21}$  the same as  $\frac{1}{3}$

.....

18. Is  $\frac{6}{9}$  the same as  $\frac{8}{12}$

.....

19. Is  $\frac{5}{8}$  the same as  $\frac{10}{16}$

.....

20. Is  $\frac{4}{7}$  the same as  $\frac{8}{15}$

.....

21. Is  $\frac{9}{15}$  the same as  $\frac{12}{20}$

.....

22. Is  $\frac{7}{9}$  the same as  $\frac{14}{17}$

.....



# Fractions Equal to One

Any fraction with matching top and bottom is just 1 wearing a costume.

## Warm-Up 12 ITEMS • RECALL

1  $5/5$  as a whole number2  $8/8$ 3  $12/12$ 4  $4/4$ 5  $100/100$ 6  $2/2$ 7  $7/7$  as a whole number8  $9/9$ 9  $6/6$ 10  $20/20$ 11  $3/3$ 12  $50/50$ 

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $6/3$  as a whole number14.  $12/4$ 15.  $20/5$ 

16. How many quarters make 2

17. How many thirds make 3

18.  $10/5$  as a whole number19.  $18/6$ 20.  $24/8$ 21.  $9/2$  as a mixed number — the whole part22.  $13/5$  — the whole part



## Working space



## Error journal

# Equivalence Wall

Build the strips and find every pair that lines up.

## ✦ Warm-Up 4 ITEMS • RECALL

1 Halves in a whole

\_\_\_\_\_

2 Quarters in a whole

\_\_\_\_\_

3 Sixths in a whole

\_\_\_\_\_

4 Quarters that line up with  $\frac{1}{2}$  — type as a/b

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

5. Sixths that line up with  $\frac{1}{3}$

.....

6.  $\frac{1}{2} = ?/12$

.....

7.  $\frac{2}{3} = ?/12$

.....

8. Twelfths that line up with  $\frac{3}{4}$  — type as a/b

.....



# Week 2 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

## 2.1 • Same Number, New Clothes

OUT OF 21

Warm-Up • 1-12 • 2 | 3 | 2 | 2 | 4 | 6 | 4 | 4 | 3 | 8 | 9 | 4

Core • 13-21 • 6 | 9 | 15 | 12 | 70 | 20 | 15 | 9 | 15

Challenge • 22-25 •  $\frac{3}{4}$  | 7 |  $\frac{3}{4}$  | 8

## 2.2 • Simplest Form

OUT OF 20

Warm-Up • 1-11 •  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{1}{3}$  |  $\frac{1}{2}$  |  $\frac{1}{3}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{1}{3}$  |  $\frac{4}{5}$  |  $\frac{3}{4}$

Core • 12-20 •  $\frac{3}{4}$  |  $\frac{2}{3}$  |  $\frac{3}{4}$  |  $\frac{3}{4}$  |  $\frac{4}{5}$  |  $\frac{3}{4}$  |  $\frac{2}{3}$  |  $\frac{2}{5}$  |  $\frac{3}{4}$

Challenge • 21-24 •  $\frac{3}{4}$  | 12 |  $\frac{3}{4}$  | 15

## 2.3 • Spot the Equivalent

OUT OF 22

Warm-Up • 1-12 • yes | yes | no | yes | yes | no | yes | yes | no | yes | yes | no

Core • 13-22 • yes | no | yes |  $\frac{5}{10}$  | yes | yes | yes | no | yes | no

Challenge • 23-26 • 9 | yes | 6 | yes

## 2.4 • Fractions Equal to One

OUT OF 22

Warm-Up • 1-12 • 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1

Core • 13-22 • 2 | 3 | 4 | 8 | 9 | 2 | 3 | 3 | 4 | 2

Challenge • 23-26 • 3 | 2 | 5 | 3

## Fri • Equivalence Wall

OUT OF 8

Warm-Up • 1-4 • 2 | 4 | 6 | 10

Core • 5-8 •  $\frac{2}{4}$  |  $\frac{2}{6}$  | 6 | 8

Challenge • 9-14 •  $\frac{9}{12}$  |  $\frac{8}{12}$  | 12 | 12 |  $\frac{2}{3}$  | 7

# Common Denominators

Cut both into the same size pieces, then just compare the counts.

## Warm-Up 11 ITEMS • RECALL

1 Common denominator of  $\frac{1}{2}$  and  $\frac{1}{4}$

\_\_\_\_\_

2 Common denominator of  $\frac{1}{3}$  and  $\frac{1}{6}$

\_\_\_\_\_

3 Common denominator of  $\frac{1}{2}$  and  $\frac{1}{3}$

\_\_\_\_\_

4  $\frac{1}{2}$  as quarters — the numerator

\_\_\_\_\_

5  $\frac{1}{3}$  as sixths — the numerator

\_\_\_\_\_

6 Common denominator of  $\frac{1}{4}$  and  $\frac{1}{8}$

\_\_\_\_\_

7 Common denominator of  $\frac{1}{2}$  and  $\frac{1}{6}$

\_\_\_\_\_

8 Common denominator of  $\frac{1}{3}$  and  $\frac{1}{9}$

\_\_\_\_\_

9  $\frac{1}{2}$  as sixths — the numerator

\_\_\_\_\_

10  $\frac{1}{3}$  as ninths — the numerator

\_\_\_\_\_

11  $\frac{1}{4}$  as eighths — the numerator

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Common denominator of  $\frac{2}{3}$  and  $\frac{3}{4}$

\_\_\_\_\_

13. Common denominator of  $\frac{3}{5}$  and  $\frac{1}{2}$

\_\_\_\_\_

14.  $\frac{2}{3}$  as twelfths — the numerator

\_\_\_\_\_

15.  $\frac{3}{4}$  as twelfths — the numerator

\_\_\_\_\_

16. Common denominator of  $\frac{5}{6}$  and  $\frac{3}{8}$

\_\_\_\_\_

17. Smallest common denominator of  $\frac{1}{4}$  and  $\frac{1}{6}$

\_\_\_\_\_

18. Of  $\frac{1}{3}$  and  $\frac{1}{5}$

\_\_\_\_\_

19. Of  $\frac{2}{5}$  and  $\frac{3}{10}$

\_\_\_\_\_

20. Of  $\frac{3}{8}$  and  $\frac{1}{6}$

\_\_\_\_\_

21. Of  $\frac{5}{12}$  and  $\frac{1}{8}$

\_\_\_\_\_



**★ Challenge** NOT GRADED • REASON IT OUT

22. Smallest common denominator of  $\frac{1}{4}$ ,  $\frac{1}{6}$  and  $\frac{1}{8}$

23. Common denominator of  $\frac{7}{10}$  and  $\frac{2}{15}$

24. Smallest common denominator of  $\frac{1}{3}$ ,  $\frac{1}{4}$  and  $\frac{1}{6}$

25. Common denominator of  $\frac{5}{9}$  and  $\frac{7}{12}$

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Benchmarks

Nearer 0, a half, or 1? Decide before you calculate anything.

## Warm-Up 12 ITEMS • RECALL

1 Is  $\frac{1}{8}$  nearer 0 or 1 — type 0 or 1

\_\_\_\_\_

2 Is  $\frac{7}{8}$  nearer 0 or 1

\_\_\_\_\_

3  $\frac{4}{8}$  simplified — type as a/b

\_\_\_\_\_

4 Is  $\frac{5}{6}$  nearer  $\frac{1}{2}$  or 1 — type  $\frac{1}{2}$  or 1

\_\_\_\_\_

5 Is  $\frac{1}{5}$  nearer 0 or  $\frac{1}{2}$  — type 0 or  $\frac{1}{2}$

\_\_\_\_\_

6 Is  $\frac{3}{5}$  more or less than  $\frac{1}{2}$  — type more or less

\_\_\_\_\_

7 Is  $\frac{1}{10}$  nearer 0 or 1 — type 0 or 1

\_\_\_\_\_

8 Is  $\frac{9}{10}$  nearer 0 or 1

\_\_\_\_\_

9  $\frac{5}{10}$  simplified — type as a/b

\_\_\_\_\_

10 Is  $\frac{1}{6}$  nearer 0 or  $\frac{1}{2}$  — type 0 or  $\frac{1}{2}$

\_\_\_\_\_

11 Is  $\frac{3}{5}$  nearer  $\frac{1}{2}$  or 1 — type  $\frac{1}{2}$  or 1

\_\_\_\_\_

12  $\frac{6}{12}$  simplified — type as a/b

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is  $\frac{5}{9}$  more or less than  $\frac{1}{2}$

.....

14. Is  $\frac{4}{9}$  more or less than  $\frac{1}{2}$

.....

15. Is  $\frac{7}{16}$  more or less than  $\frac{1}{2}$

.....

16. Is  $\frac{9}{16}$  more or less than  $\frac{1}{2}$

.....

17. Nearest benchmark to  $\frac{11}{12}$  — type 0,  $\frac{1}{2}$  or 1

.....

18. Is  $\frac{11}{20}$  more or less than  $\frac{1}{2}$

.....

19. Nearest benchmark to  $\frac{9}{10}$  — type 0,  $\frac{1}{2}$  or 1

.....



## Working space



## Error journal

# Comparing & Ordering

Common denominator, or reason from benchmarks. Say which you used.

## Warm-Up 12 ITEMS • RECALL

1 Larger:  $\frac{1}{2}$  or  $\frac{1}{3}$  — type it

2 Larger:  $\frac{3}{4}$  or  $\frac{1}{2}$

3 Larger:  $\frac{1}{4}$  or  $\frac{1}{8}$

4 Larger:  $\frac{2}{3}$  or  $\frac{1}{3}$

5 Smaller:  $\frac{1}{5}$  or  $\frac{1}{6}$

6 Larger:  $\frac{5}{8}$  or  $\frac{3}{8}$

7 Larger:  $\frac{1}{2}$  or  $\frac{1}{4}$  — type it

8 Larger:  $\frac{2}{3}$  or  $\frac{1}{2}$

9 Larger:  $\frac{3}{5}$  or  $\frac{2}{5}$

10 Larger:  $\frac{1}{3}$  or  $\frac{1}{4}$

11 Smaller:  $\frac{1}{8}$  or  $\frac{1}{10}$

12 Larger:  $\frac{5}{6}$  or  $\frac{1}{2}$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Larger:  $\frac{2}{3}$  or  $\frac{3}{4}$  — type it

14. Larger:  $\frac{3}{5}$  or  $\frac{5}{8}$

15. Larger:  $\frac{5}{6}$  or  $\frac{7}{9}$

16. Smallest of  $\frac{1}{3}$ ,  $\frac{2}{5}$ ,  $\frac{1}{4}$  — type it

17. Largest of  $\frac{2}{3}$ ,  $\frac{5}{8}$ ,  $\frac{7}{12}$

18. Larger:  $\frac{2}{3}$  or  $\frac{5}{8}$  — type it

19. Larger:  $\frac{3}{7}$  or  $\frac{1}{2}$

20. Larger:  $\frac{7}{12}$  or  $\frac{5}{9}$

21. Larger:  $\frac{4}{5}$  or  $\frac{7}{9}$

22. Smaller:  $\frac{5}{12}$  or  $\frac{3}{8}$



## Working space



## Error journal

# On the Number Line

Every fraction has one address between 0 and 1. Find it.

## Warm-Up 10 ITEMS • RECALL

1 Halfway between 0 and 1  
— type as a/b

\_\_\_\_\_

2 Halfway between 0 and  $\frac{1}{2}$

\_\_\_\_\_

3 Halfway between  $\frac{1}{2}$  and 1

\_\_\_\_\_

4 Marks between 0 and 1 if  
split into quarters

\_\_\_\_\_

5 Marks if split into fifths

\_\_\_\_\_

6 Halfway between 0 and  $\frac{1}{4}$  — type as a/b

\_\_\_\_\_

7 Halfway between  $\frac{1}{4}$  and  $\frac{1}{2}$

\_\_\_\_\_

8 Halfway between  $\frac{3}{4}$  and 1

\_\_\_\_\_

9 Marks between 0 and 1 if  
split into fifths

\_\_\_\_\_

10 Halfway between  $\frac{1}{4}$   
and  $\frac{1}{2}$  — type as a/b

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11.  $\frac{1}{2}$  as a decimal

\_\_\_\_\_

12.  $\frac{3}{4}$  as a decimal

\_\_\_\_\_

13.  $\frac{1}{4}$  as a decimal

\_\_\_\_\_

14.  $\frac{1}{5}$  as a decimal

\_\_\_\_\_

15.  $\frac{3}{5}$  as a decimal

\_\_\_\_\_

16.  $\frac{7}{8}$  as a decimal

\_\_\_\_\_

17. A fraction exactly between  $\frac{1}{3}$  and  $\frac{2}{3}$  — type as a/b

\_\_\_\_\_



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

## ✦ Warm-Up 4 ITEMS • RECALL

1  $1/2 = ?/8$  — the numerator

\_\_\_\_\_

2  $4/8$  simplified — type as a/b

\_\_\_\_\_

3  $1/3 = ?/9$  — the numerator

\_\_\_\_\_

4  $6/8$  simplified — type as a/b

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.

5.  $6/8$  simplified

\_\_\_\_\_

6. Common denominator of  $2/3$  and  $3/4$

\_\_\_\_\_

7. Larger:  $3/5$  or  $5/8$  — type it

\_\_\_\_\_

8. Is  $5/9$  more or less than  $1/2$

\_\_\_\_\_

9. Larger:  $2/3$  or  $5/8$  — type it

\_\_\_\_\_

10. Smallest common denominator of  $1/4$  and  $1/6$

\_\_\_\_\_

11.  $3/4$  as a decimal

\_\_\_\_\_





## Working space



## Error journal

# Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 3.1 • Common Denominators

OUT OF 21

Warm-Up • 1-11 • 4 | 6 | 6 | 2 | 2 | 8 | 6 | 9 | 3 | 3 | 2

Core • 12-21 • 12 | 10 | 8 | 9 | 24 | 12 | 15 | 10 | 24 | 24

Challenge • 22-25 • 24 | 30 | 12 | 36

## 3.2 • Benchmarks

OUT OF 19

Warm-Up • 1-12 • 0 | 1 |  $\frac{1}{2}$  | 1 | 0 | more | 0 | 1 |  $\frac{1}{2}$  | 0 |  $\frac{1}{2}$  |  $\frac{1}{2}$

Core • 13-19 • more | less | less | more | 1 | more | 1

Challenge • 20-23 • more |  $\frac{1}{2}$  | less |  $\frac{1}{2}$

## 3.3 • Comparing & Ordering

OUT OF 22

Warm-Up • 1-12 •  $\frac{1}{2}$  |  $\frac{3}{4}$  |  $\frac{1}{4}$  |  $\frac{2}{3}$  |  $\frac{1}{6}$  |  $\frac{5}{8}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{3}{5}$  |  $\frac{1}{3}$  |  $\frac{1}{10}$  |  $\frac{5}{6}$

Core • 13-22 •  $\frac{3}{4}$  |  $\frac{5}{8}$  |  $\frac{5}{6}$  |  $\frac{1}{4}$  |  $\frac{2}{3}$  |  $\frac{2}{3}$  |  $\frac{1}{2}$  |  $\frac{7}{12}$  |  $\frac{4}{5}$  |  $\frac{3}{8}$

Challenge • 23-26 •  $\frac{7}{8}$  |  $\frac{11}{15}$  |  $\frac{7}{10}$  |  $\frac{2}{3}$

## 3.4 • On the Number Line

OUT OF 10

Warm-Up • 1-9 •  $\frac{1}{2}$  |  $\frac{1}{4}$  |  $\frac{3}{4}$  | 3 | 4 |  $\frac{1}{8}$  |  $\frac{3}{8}$  |  $\frac{7}{8}$  | 4

Core • 10-10 •  $\frac{3}{8}$

Challenge • 11-21 • 0.5 | 0.75 | 0.25 | 0.2 | 0.6 | 0.875 |  $\frac{1}{2}$  | 0.125 | 0.1 | 0.375 |  $\frac{1}{2}$

## Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 4 |  $\frac{1}{2}$  | 3 |  $\frac{3}{4}$

Core • 5-11 •  $\frac{3}{4}$  | 12 |  $\frac{5}{8}$  | more | 0.75 |  $\frac{2}{3}$  | 12

Challenge • 12-15 •  $\frac{7}{8}$  | 0.375 | less |  $\frac{3}{4}$

# Adding Like Denominators

Same size pieces already. Add the counts, leave the bottom alone.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{3}{10} + \frac{2}{10}$

2  $\frac{1}{10} + \frac{4}{10}$

3  $\frac{1}{12} + \frac{5}{12}$

4  $\frac{5}{12} + \frac{3}{12}$

5  $\frac{7}{10} + \frac{1}{10}$

6  $\frac{4}{15} + \frac{6}{15}$

7  $\frac{7}{20} + \frac{3}{20}$

8  $\frac{3}{14} + \frac{4}{14}$

9  $\frac{1}{4} + \frac{1}{4}$  — type as a/b

10  $\frac{1}{5} + \frac{2}{5}$

11  $\frac{1}{8} + \frac{3}{8}$

12  $\frac{2}{6} + \frac{1}{6}$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $\frac{1}{3} + \frac{1}{3}$

14.  $\frac{3}{8} + \frac{3}{8}$

15.  $\frac{2}{9} + \frac{4}{9}$

16.  $\frac{5}{6} + \frac{1}{6}$

17.  $\frac{7}{8} + \frac{5}{8}$  — type as a/b

18.  $\frac{3}{5} + \frac{4}{5}$

19.  $\frac{1}{6} + \frac{1}{6}$  — type as a/b

20.  $\frac{2}{7} + \frac{3}{7}$

21.  $\frac{3}{9} + \frac{3}{9}$



# Subtracting Like Denominators

Same move, other direction.

## Warm-Up 10 ITEMS • RECALL

1  $9/10 - 4/10$

2  $8/10 - 3/10$

3  $1 - 1/4$

4  $1 - 3/8$

5  $11/12 - 5/12$

6  $1 - 2/5$

7  $7/12 - 1/12$

8  $11/15 - 1/15$

9  $13/20 - 3/20$

10  $3/4 - 1/4$  — type as a/b

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11.  $4/5 - 1/5$

12.  $7/8 - 3/8$

13.  $5/6 - 1/6$

14.  $2/3 - 1/3$

15.  $7/9 - 1/9$

16.  $2 - 3/4$  — type as a/b

17.  $1 \frac{1}{2} - 3/4$  — type as a/b

18.  $5/8 - 1/8$  — type as a/b

19.  $5/7 - 2/7$

20.  $4/6 - 1/6$



# Unlike Denominators

The grade-five ceiling. Convert first, then add.

## Warm-Up 12 ITEMS • RECALL

1 A whole cut into 4 equal parts — one part is 1 over what

\_\_\_\_\_

2 How many quarters make one whole

\_\_\_\_\_

3 How many sixths make one whole

\_\_\_\_\_

4  $\frac{2}{4}$  in its simplest form — the numerator

\_\_\_\_\_

5 Is  $\frac{3}{4}$  more or less than a half

\_\_\_\_\_

6  $\frac{1}{2} = \frac{?}{4}$  — give the numerator

\_\_\_\_\_

7  $\frac{1}{3} = \frac{?}{6}$  — give the numerator

\_\_\_\_\_

8  $\frac{1}{4} = \frac{?}{8}$  — give the numerator

\_\_\_\_\_

9  $\frac{2}{3} = \frac{?}{6}$  — give the numerator

\_\_\_\_\_

10  $\frac{3}{4} = \frac{?}{8}$  — give the numerator

\_\_\_\_\_

11  $\frac{2}{4}$  in its simplest form — give the numerator

\_\_\_\_\_

12  $\frac{3}{6}$  in its simplest form — give the numerator

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $\frac{4}{8}$  in its simplest form — give the numerator

.....

14.  $\frac{2}{6}$  in its simplest form — give the numerator

.....

15. How many 2ths equal one half

.....

16. How many 4ths equal one half

.....

17. How many 8ths equal one half

.....

18. Is  $\frac{1}{2}$  more or less than a half — answer more, less or equal

.....

19. Is  $\frac{1}{4}$  more or less than a half — answer more, less or equal

.....

20. Is  $\frac{3}{4}$  more or less than a half — answer more, less or equal

.....

21. Is  $\frac{2}{3}$  more or less than a half — answer more, less or equal

.....

22. On a number line cut into 2ths, which mark is one whole

.....





# Fractions in Words

Read it twice, then decide whether it is an add or a subtract.

## Warm-Up 12 ITEMS • RECALL

1 Ate  $\frac{1}{4}$  of a pizza, then  $\frac{1}{4}$  more — total, as a/b

\_\_\_\_\_

2 Had  $\frac{3}{4}$ , ate  $\frac{1}{4}$  — left

\_\_\_\_\_

3  $\frac{1}{2}$  hour plus  $\frac{1}{2}$  hour — hours

\_\_\_\_\_

4  $\frac{2}{5}$  of a jug plus  $\frac{1}{5}$  — as a/b

\_\_\_\_\_

5 1 whole minus  $\frac{1}{3}$  — as a/b

\_\_\_\_\_

6 Drank  $\frac{1}{3}$  of a bottle, then  $\frac{1}{3}$  more — total, as a/b

\_\_\_\_\_

7 Had  $\frac{5}{6}$ , used  $\frac{1}{6}$  — left

\_\_\_\_\_

8  $\frac{1}{4}$  hour plus  $\frac{3}{4}$  hour — hours

\_\_\_\_\_

9  $\frac{3}{8}$  of a tank plus  $\frac{1}{8}$  — as a/b

\_\_\_\_\_

10 Read  $\frac{1}{2}$  a book, then  $\frac{1}{4}$  — as a/b

\_\_\_\_\_

11 Had  $\frac{7}{8}$ , gave away  $\frac{3}{8}$  — as a/b

\_\_\_\_\_

12 Walked  $\frac{1}{2}$  mile then  $\frac{1}{4}$  mile — total, as a/b

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. A  $\frac{3}{4}$  cup recipe, you have  $\frac{1}{2}$  — how much more, as a/b

\_\_\_\_\_

14. Read  $\frac{2}{3}$  of a book, then  $\frac{1}{6}$  more — total, as a/b

\_\_\_\_\_

15. Ran  $\frac{5}{6}$  km and walked  $\frac{1}{3}$  km — total, as a/b

\_\_\_\_\_

16. 1 hour minus  $\frac{1}{4}$  hour — minutes

\_\_\_\_\_

17. Walked  $\frac{2}{5}$  km then  $\frac{1}{2}$  km — total as a/b

\_\_\_\_\_

18.  $\frac{3}{4}$  of an hour in minutes

\_\_\_\_\_

19.  $\frac{1}{3}$  of an hour in minutes

\_\_\_\_\_

20. Ate  $\frac{3}{8}$  of a pie, dad ate  $\frac{1}{4}$  — total as a/b

\_\_\_\_\_

21.  $1 - \frac{5}{8}$  as a/b

\_\_\_\_\_

22.  $\frac{1}{8} + \frac{1}{8}$  — as a/b

\_\_\_\_\_



## Working space



## Error journal

# Closest to One

Draw cards, build a fraction, get nearest to 1 without going over.

## ✦ Warm-Up 4 ITEMS • RECALL

1 Which is nearer 1:  $\frac{3}{4}$  or  $\frac{1}{2}$  — type it

\_\_\_\_\_

2 Which is nearer 1:  $\frac{7}{8}$  or  $\frac{3}{4}$

\_\_\_\_\_

3 Which is nearer 1:  $\frac{5}{6}$  or  $\frac{2}{3}$  — type it

\_\_\_\_\_

4 Which is nearer 1:  $\frac{9}{10}$  or  $\frac{4}{5}$

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

5. From digits 5 and 6, the fraction under 1 — type as a/b

.....

6. From 7 and 8, the fraction under 1

.....

7. Which is nearer 1:  $\frac{7}{8}$  or  $\frac{8}{9}$  — type it

.....

8. Which is nearer 1:  $\frac{11}{12}$  or  $\frac{5}{6}$

.....



## SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 4.1 • Adding Like Denominators

OUT OF 8

Warm-Up • 1-3 •  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{1}{2}$

Core • 4-8 •  $\frac{2}{3}$  |  $\frac{4}{5}$  |  $\frac{2}{3}$  |  $\frac{1}{2}$  |  $\frac{1}{2}$

Challenge • 9-25 •  $\frac{1}{2}$  |  $\frac{3}{5}$  |  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{3}{4}$  |  $\frac{2}{3}$  | 1 |  $\frac{3}{2}$  |  $\frac{7}{5}$  |  $\frac{1}{3}$  |  $\frac{5}{7}$  |  $\frac{2}{3}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{5}{3}$  |  $\frac{8}{5}$

## 4.2 • Subtracting Like Denominators

OUT OF 9

Warm-Up • 1-2 •  $\frac{1}{2}$  |  $\frac{1}{2}$

Core • 3-9 •  $\frac{3}{4}$  |  $\frac{5}{8}$  |  $\frac{1}{2}$  |  $\frac{3}{5}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{1}{2}$

Challenge • 10-24 •  $\frac{1}{2}$  |  $\frac{3}{5}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{1}{3}$  |  $\frac{2}{3}$  |  $\frac{5}{4}$  |  $\frac{3}{4}$  |  $\frac{1}{2}$  |  $\frac{3}{7}$  |  $\frac{1}{2}$  |  $\frac{1}{3}$  |  $\frac{5}{8}$  |  $\frac{7}{4}$  |  $\frac{3}{2}$

## 4.3 • Unlike Denominators

OUT OF 26

Warm-Up • 1-5 • 4 | 4 | 6 | 1 | more

Core • 6-26 • 2 | 2 | 2 | 4 | 6 | 1 | 1 | 1 | 1 | 1 | 2 | 4 | equal | less | more | more | 2 | 3 | 4 | 6 | 8

Challenge • 27-52 •  $\frac{3}{4}$  |  $\frac{1}{2}$  |  $\frac{5}{8}$  |  $\frac{3}{8}$  |  $\frac{1}{4}$  |  $\frac{1}{2}$  |  $\frac{5}{6}$  |  $\frac{7}{8}$  |  $\frac{11}{12}$  |  $\frac{5}{12}$  |  $\frac{1}{3}$  |  $\frac{17}{12}$  | 1 |  $\frac{2}{3}$  |  $\frac{3}{4}$  |  $\frac{4}{9}$  |  $\frac{3}{10}$  |  $\frac{5}{8}$  |  $\frac{1}{3}$  |  $\frac{5}{6}$  |  $\frac{7}{10}$  |  $\frac{3}{8}$  |  $\frac{1}{12}$  |  $\frac{13}{20}$  |  $\frac{19}{12}$  |  $\frac{7}{8}$

## 4.4 • Fractions in Words

OUT OF 21

Warm-Up • 1-11 •  $\frac{1}{2}$  |  $\frac{1}{2}$  | 1 |  $\frac{3}{5}$  |  $\frac{2}{3}$  |  $\frac{2}{3}$  |  $\frac{2}{3}$  | 1 |  $\frac{1}{2}$  |  $\frac{3}{4}$  |  $\frac{1}{2}$

Core • 12-21 •  $\frac{3}{4}$  |  $\frac{1}{4}$  |  $\frac{5}{6}$  |  $\frac{7}{6}$  | 45 |  $\frac{9}{10}$  | 45 | 20 |  $\frac{5}{8}$  |  $\frac{3}{8}$

Challenge • 22-26 •  $\frac{1}{4}$  |  $\frac{9}{8}$  | 40 |  $\frac{8}{5}$  | 50

## Fri • Closest to One

OUT OF 8

Warm-Up • 1-4 •  $\frac{3}{4}$  |  $\frac{7}{8}$  |  $\frac{5}{6}$  |  $\frac{9}{10}$

Core • 5-8 •  $\frac{5}{6}$  |  $\frac{7}{8}$  |  $\frac{8}{9}$  |  $\frac{11}{12}$

Challenge • 9-14 •  $\frac{9}{10}$  |  $\frac{1}{10}$  |  $\frac{5}{6}$  |  $\frac{7}{8}$  |  $\frac{4}{5}$  |  $\frac{1}{9}$

# Fraction of a Set

Divide by the bottom, multiply by the top.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{1}{2}$  of 10  
\_\_\_\_\_2  $\frac{1}{3}$  of 12  
\_\_\_\_\_3  $\frac{1}{4}$  of 20  
\_\_\_\_\_4  $\frac{1}{5}$  of 25  
\_\_\_\_\_5  $\frac{1}{2}$  of 30  
\_\_\_\_\_6  $\frac{1}{6}$  of 30  
\_\_\_\_\_7  $\frac{1}{2}$  of 14  
\_\_\_\_\_8  $\frac{1}{3}$  of 15  
\_\_\_\_\_9  $\frac{1}{4}$  of 24  
\_\_\_\_\_10  $\frac{1}{5}$  of 35  
\_\_\_\_\_11  $\frac{1}{6}$  of 36  
\_\_\_\_\_12  $\frac{1}{8}$  of 40  
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

13.  $\frac{2}{5}$  of 30  
.....14.  $\frac{3}{4}$  of 20  
.....15.  $\frac{2}{3}$  of 18  
.....16.  $\frac{5}{6}$  of 24  
.....17.  $\frac{3}{8}$  of 40  
.....18.  $\frac{3}{4}$  of 32  
.....



# Of Money and Time

The same operator, applied to dollars and minutes.

## ✦ Warm-Up 9 ITEMS • RECALL

1  $\frac{1}{2}$  of \$20

\_\_\_\_\_

2  $\frac{1}{4}$  of \$40

\_\_\_\_\_

3  $\frac{1}{2}$  of an hour in minutes

\_\_\_\_\_

4  $\frac{1}{4}$  of an hour in minutes

\_\_\_\_\_

5  $\frac{1}{3}$  of \$30

\_\_\_\_\_

6  $\frac{1}{5}$  of \$50

\_\_\_\_\_

7  $\frac{1}{2}$  of \$10

\_\_\_\_\_

8  $\frac{1}{4}$  of \$20

\_\_\_\_\_

9  $\frac{1}{5}$  of \$25

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10.  $\frac{3}{4}$  of \$60

.....

11.  $\frac{2}{3}$  of an hour in minutes

.....

12.  $\frac{3}{5}$  of \$45

.....

13.  $\frac{5}{6}$  of an hour in minutes

.....

14.  $\frac{2}{5}$  of \$75

.....

15.  $\frac{2}{3}$  of \$45

.....

16.  $\frac{3}{4}$  of an hour in minutes

.....

17.  $\frac{2}{5}$  of \$50

.....





# Working Backwards

Given the part, find the whole.

## Warm-Up 12 ITEMS • RECALL

1 Half a number is 6. The number  
\_\_\_\_\_

2 A third is 4. The number  
\_\_\_\_\_

3 A quarter is 5. The number  
\_\_\_\_\_

4 A fifth is 3. The number  
\_\_\_\_\_

5 Half is 15. The number  
\_\_\_\_\_

6 A sixth is 2. The number  
\_\_\_\_\_

7  $\frac{1}{2}$  of 100  
\_\_\_\_\_

8  $\frac{1}{4}$  of 100  
\_\_\_\_\_

9  $\frac{1}{10}$  of 100  
\_\_\_\_\_

10  $\frac{1}{5}$  of 100  
\_\_\_\_\_

11  $\frac{3}{4}$  of 100  
\_\_\_\_\_

12  $\frac{1}{2}$  of 60  
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $\frac{2}{3}$  of a number is 12. The number  
.....

14.  $\frac{3}{4}$  of a number is 15. The number  
.....

15.  $\frac{2}{5}$  of a number is 10. The number  
.....

16.  $\frac{5}{6}$  of a number is 20. The number  
.....

17.  $\frac{3}{8}$  of a number is 9. The number  
.....

18.  $\frac{2}{5}$  of 200  
.....

19.  $\frac{3}{10}$  of 150  
.....

20.  $\frac{5}{8}$  of 160  
.....

21.  $\frac{2}{3}$  of 150  
.....

22.  $\frac{4}{5}$  of 250  
.....



# Mixed Fraction Problems

No signposting. Read, decide, then compute.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{1}{2}$  of 16  
\_\_\_\_\_2  $1 - \frac{1}{2}$  — as a/b  
\_\_\_\_\_3  $\frac{2}{6}$  simplified — as a/b  
\_\_\_\_\_4  $\frac{1}{3}$  of 15  
\_\_\_\_\_5  $\frac{1}{2}$  of 8  
\_\_\_\_\_6  $\frac{1}{3}$  of 9  
\_\_\_\_\_7  $\frac{1}{4}$  of 16  
\_\_\_\_\_8  $\frac{1}{2}$  of 22  
\_\_\_\_\_9  $\frac{1}{5}$  of 20  
\_\_\_\_\_10  $\frac{1}{6}$  of 24  
\_\_\_\_\_11  $\frac{3}{4}$  of 24  
\_\_\_\_\_12  $\frac{2}{5}$  of a number is 8.  
The number  
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Larger:  $\frac{5}{8}$  or  $\frac{2}{3}$  — type it  
.....14.  $\frac{3}{4}$  of 28  
.....15.  $\frac{2}{3}$  of 33  
.....16.  $\frac{4}{5}$  of 45  
.....17.  $\frac{5}{8}$  of 48  
.....18.  $\frac{3}{7}$  of 42  
.....19.  $\frac{1}{4} + \frac{1}{4}$  — type as a/b  
.....20.  $\frac{3}{4} - \frac{1}{4}$  — as a/b  
.....21.  $\frac{2}{3} + \frac{1}{6}$  — as a/b  
.....22.  $\frac{5}{8} - \frac{1}{4}$  — as a/b  
.....



## Working space



## Error journal

# Fold the Universe, Built

Finish the paper number line with every fraction landed on it.

NAME \_\_\_\_\_

DATE \_\_\_\_\_

RIGHT \_\_\_\_ OF \_\_\_\_ TRIED

**Warm-Up** 4 ITEMS • RECALL

- 1 Folds to get halves  
\_\_\_\_\_
- 2 Folds to get quarters  
\_\_\_\_\_
- 3  $\frac{1}{2}$  of 12  
\_\_\_\_\_
- 4  $\frac{1}{4}$  of 12  
\_\_\_\_\_

**Core** GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

5. Folds to get eighths  
.....
6. Sections after 4 folds  
.....
7.  $\frac{2}{3}$  of 24  
.....
8.  $\frac{3}{5}$  of 40  
.....



## SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 5.1 • Fraction of a Set

OUT OF 18

Warm-Up • 1-12 • 5 | 4 | 5 | 5 | 15 | 5 | 7 | 5 | 6 | 7 | 6 | 5

Core • 13-18 • 12 | 15 | 12 | 20 | 15 | 24

Challenge • 19-22 • 30 | 84 | 51 | 105

## 5.2 • Of Money and Time

OUT OF 17

Warm-Up • 1-9 • 10 | 10 | 30 | 15 | 10 | 10 | 5 | 5 | 5

Core • 10-17 • 45 | 40 | 27 | 50 | 30 | 30 | 45 | 20

Challenge • 18-19 • 36 | 35

## 5.3 • Working Backwards

OUT OF 22

Warm-Up • 1-12 • 12 | 12 | 20 | 15 | 30 | 12 | 50 | 25 | 10 | 20 | 75 | 30

Core • 13-22 • 18 | 20 | 25 | 24 | 24 | 80 | 45 | 100 | 100 | 200

Challenge • 23-26 • 63 | 50 | 140 | 60

## 5.4 • Mixed Fraction Problems

OUT OF 18

Warm-Up • 1-10 • 8 |  $\frac{1}{2}$  |  $\frac{1}{3}$  | 5 | 4 | 3 | 4 | 11 | 4 | 4

Core • 11-18 • 18 | 20 |  $\frac{2}{3}$  | 21 | 22 | 36 | 30 | 18

Challenge • 19-26 •  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{5}{6}$  |  $\frac{3}{8}$  |  $\frac{17}{12}$  | 35 | 225 | 60

## Fri • Fold the Universe, Built

OUT OF 8

Warm-Up • 1-4 • 1 | 2 | 6 | 3

Core • 5-8 • 3 | 16 | 16 | 24

Challenge • 9-14 • 64 | 5 | 15 | 84 | 110 | 183



# The Equivalence Proof

Show the same amount two different ways and say why they match.

## Warm-Up 12 ITEMS • RECALL

1  $1/2 = ?/4$  — the numerator

\_\_\_\_\_

2  $2/4$  simplified — type as a/b

\_\_\_\_\_

3  $1/2 \times 3/3$  — type as a/b

\_\_\_\_\_

4  $3/6$  simplified

\_\_\_\_\_

5  $3/3$  as a whole number

\_\_\_\_\_

6 Multiplying by 1 changes a number by how much

\_\_\_\_\_

7  $1/2$  of 16

\_\_\_\_\_

8  $3/4$  of 8

\_\_\_\_\_

9  $2/3 \times 4/4$  — type as a/b

\_\_\_\_\_

10  $8/12$  simplified

\_\_\_\_\_

11  $5/6 \times 3/3$

\_\_\_\_\_

12  $15/18$  simplified

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. What  $3/4$  was multiplied by to become  $12/16$

.....

14.  $2/3$  of 27

.....

15.  $7/10 - 1/5$

.....

16.  $18/27$  simplified — type as a/b

.....

17. Two fractions equal to  $4/6$  — type the simplest

.....

18.  $1/2 + 1/2$  — type as a/b

.....

19.  $1/4 + 1/4$

.....

20.  $2/3 - 1/3$

.....

21.  $5/8 - 1/8$

.....

22.  $1/2 + 1/3$

.....



# Number Line Defence

Every fraction has one place. Put it there and argue for it.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{1}{3}$  of 12

\_\_\_\_\_

2  $\frac{1}{2}$  of 18

\_\_\_\_\_

3  $\frac{1}{5}$  of 30

\_\_\_\_\_

4  $\frac{1}{6}$  of 42

\_\_\_\_\_

5  $\frac{3}{8}$  of 88

\_\_\_\_\_

6  $\frac{7}{12} - \frac{1}{4}$ 

\_\_\_\_\_

7  $\frac{1}{2}$  as a decimal

\_\_\_\_\_

8  $\frac{1}{4}$  as a decimal

\_\_\_\_\_

9  $\frac{3}{4}$  as a decimal

\_\_\_\_\_

10  $\frac{1}{5}$  as a decimal

\_\_\_\_\_

11  $\frac{2}{5}$  as a decimal

\_\_\_\_\_

12  $\frac{1}{10}$  as a decimal

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $\frac{3}{8}$  as a decimal

.....

14.  $\frac{5}{8}$  as a decimal

.....

15.  $\frac{7}{10}$  as a decimal

.....

16. Which sits further right:  $\frac{2}{3}$  or 0.6 — type it

.....

17.  $\frac{3}{5}$  as a decimal

.....

18.  $\frac{7}{8}$  as a decimal

.....

19. Between 0.3 and 0.4 — type  $\frac{1}{3}$  or  $\frac{2}{5}$ , whichever fits

.....

20.  $\frac{1}{4} + \frac{1}{2}$ 

.....

21.  $\frac{3}{5} - \frac{1}{5}$ 

.....

22.  $\frac{2}{5} + \frac{1}{2}$ 

.....



# Mission Review

Everything from six weeks, mixed together.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{6}{8}$  simplified — type as a/b

\_\_\_\_\_

2  $\frac{1}{2}$  of 14

\_\_\_\_\_

3  $1 - \frac{1}{3}$  — as a/b

\_\_\_\_\_

4 Larger:  $\frac{1}{2}$  or  $\frac{1}{3}$  — type it

\_\_\_\_\_

5 Half of a half — type as a/b

\_\_\_\_\_

6 Half of a quarter

\_\_\_\_\_

7 Two quarters

\_\_\_\_\_

8 Three thirds

\_\_\_\_\_

9  $\frac{1}{2}$  of 40

\_\_\_\_\_

10  $\frac{1}{4}$  of 40

\_\_\_\_\_

11  $\frac{3}{4}$  of 32

\_\_\_\_\_

12 Common denominator of  $\frac{5}{6}$  and  $\frac{3}{8}$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $\frac{2}{5}$  of a number is 10. The number

.....

14.  $\frac{1}{2}$  of  $\frac{1}{3}$  — type as a/b

.....

15.  $\frac{1}{3}$  of  $\frac{1}{4}$

.....

16.  $\frac{2}{3}$  of  $\frac{1}{2}$

.....

17.  $\frac{3}{4}$  of  $\frac{1}{2}$

.....

18.  $\frac{1}{5}$  of  $\frac{1}{2}$

.....

19.  $\frac{1}{4} + \frac{1}{4}$  — as a/b

.....

20.  $\frac{1}{2}$  as a decimal

.....

21.  $\frac{2}{3} + \frac{1}{4}$  — as a/b

.....

22.  $\frac{3}{4} - \frac{1}{3}$  — as a/b

.....



# Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{2}{4}$  simplified — type as a/b

\_\_\_\_\_

2  $\frac{1}{4}$  of 16

\_\_\_\_\_

3 Is  $\frac{3}{6}$  the same as  $\frac{1}{2}$  — yes or no

\_\_\_\_\_

4  $\frac{2}{3}$  of 9

\_\_\_\_\_

5  $\frac{1}{4}$  of 32

\_\_\_\_\_

6  $\frac{1}{3}$  of 24

\_\_\_\_\_

7  $\frac{12}{16}$  simplified

\_\_\_\_\_

8  $\frac{5}{6}$  of 24

\_\_\_\_\_

9 Larger:  $\frac{2}{3}$  or  $\frac{3}{4}$  — type it

\_\_\_\_\_

10  $\frac{5}{8}$  of 64

\_\_\_\_\_

11  $\frac{2}{3}$  of 42

\_\_\_\_\_

12  $\frac{1}{3} + \frac{1}{3}$  — as a/b

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $\frac{1}{5}$  as a decimal

.....

14.  $\frac{5}{6} - \frac{1}{6}$  — as a/b

.....

15.  $\frac{1}{2} + \frac{1}{3}$  — as a/b

.....

16.  $\frac{3}{8}$  as a decimal

.....

17.  $\frac{2}{3} + \frac{3}{4}$  — as a/b

.....

18. After spending  $\frac{3}{5}$ , \$20 is left. The starting amount

.....

19.  $\frac{1}{2} + \frac{1}{4}$

.....

20.  $\frac{5}{6} - \frac{1}{6}$

.....

21.  $\frac{3}{8} + \frac{1}{8}$

.....

22.  $\frac{3}{4} + \frac{1}{6}$

.....

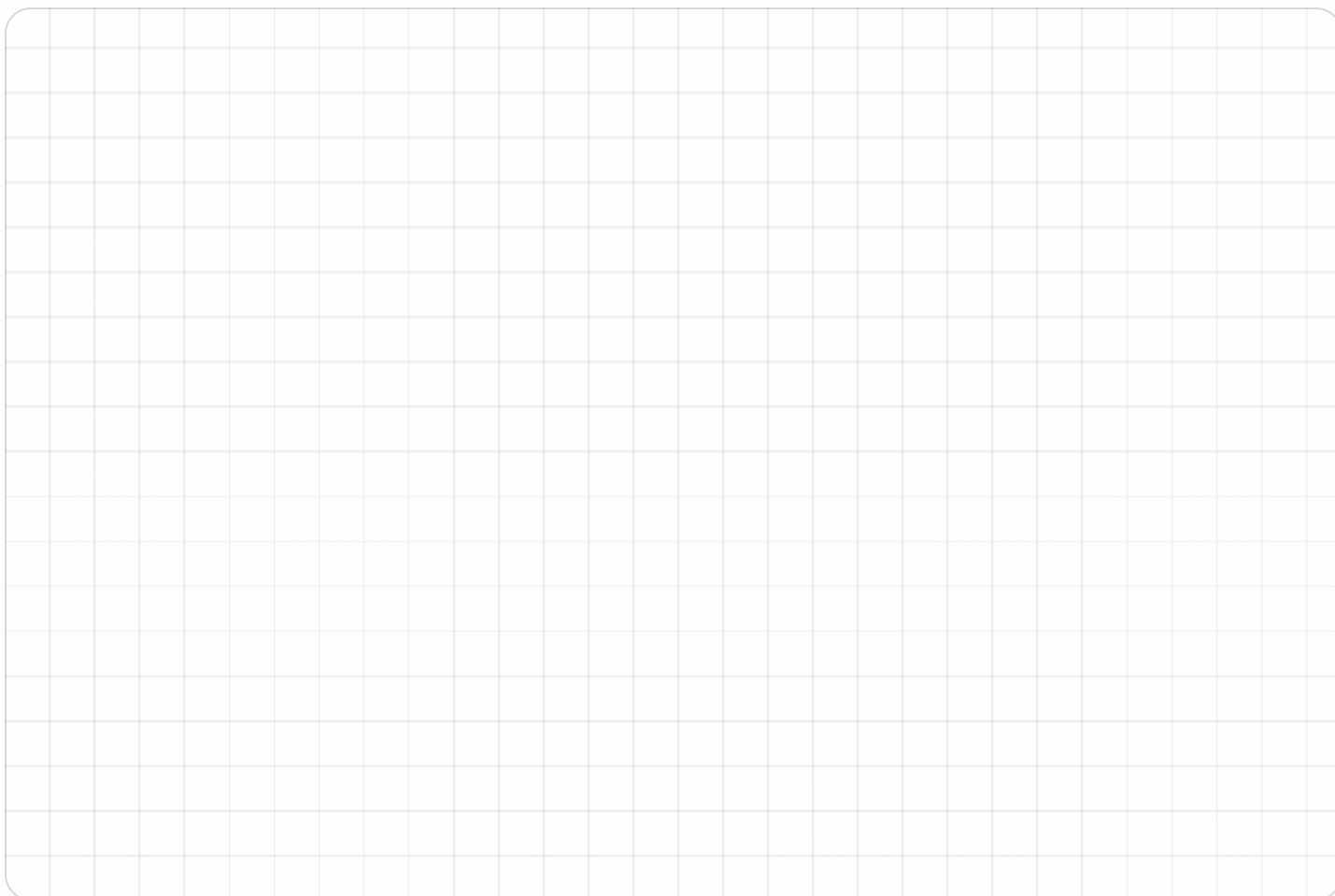
**★ Challenge** NOT GRADED • REASON IT OUT

23.  $\frac{7}{8} - \frac{1}{2}$   
.....

24.  $\frac{2}{5} + \frac{3}{10}$   
.....

25.  $\frac{5}{6} + \frac{3}{8}$   
.....

26.  $\frac{7}{9}$  of 108  
.....

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Mission 04 Test

Twelve items plus the Big Question, answered out loud.

## Warm-Up 4 ITEMS • RECALL

1  $\frac{4}{8}$  simplified — type as a/b  
\_\_\_\_\_2  $\frac{1}{2}$  of 20  
\_\_\_\_\_3  $\frac{1}{2} = \frac{?}{10}$  — the numerator  
\_\_\_\_\_4  $\frac{10}{15}$  simplified  
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.5. Larger:  $\frac{3}{5}$  or  $\frac{5}{8}$  — type it  
.....6.  $\frac{2}{3}$  of 18  
.....7. Common denominator of  $\frac{2}{3}$  and  $\frac{3}{4}$   
.....8.  $\frac{3}{4}$  of 44  
.....9.  $\frac{1}{4} + \frac{1}{4}$  — as a/b  
.....10.  $\frac{1}{2} + \frac{1}{4}$  — as a/b  
.....11.  $\frac{3}{4} - \frac{1}{3}$  — as a/b  
.....12.  $\frac{3}{4}$  as a decimal  
.....13.  $\frac{2}{3} + \frac{3}{4}$  — as a/b  
.....14.  $\frac{3}{8}$  of a number is 9. The number  
.....

**★ Challenge** NOT GRADED • REASON IT OUT

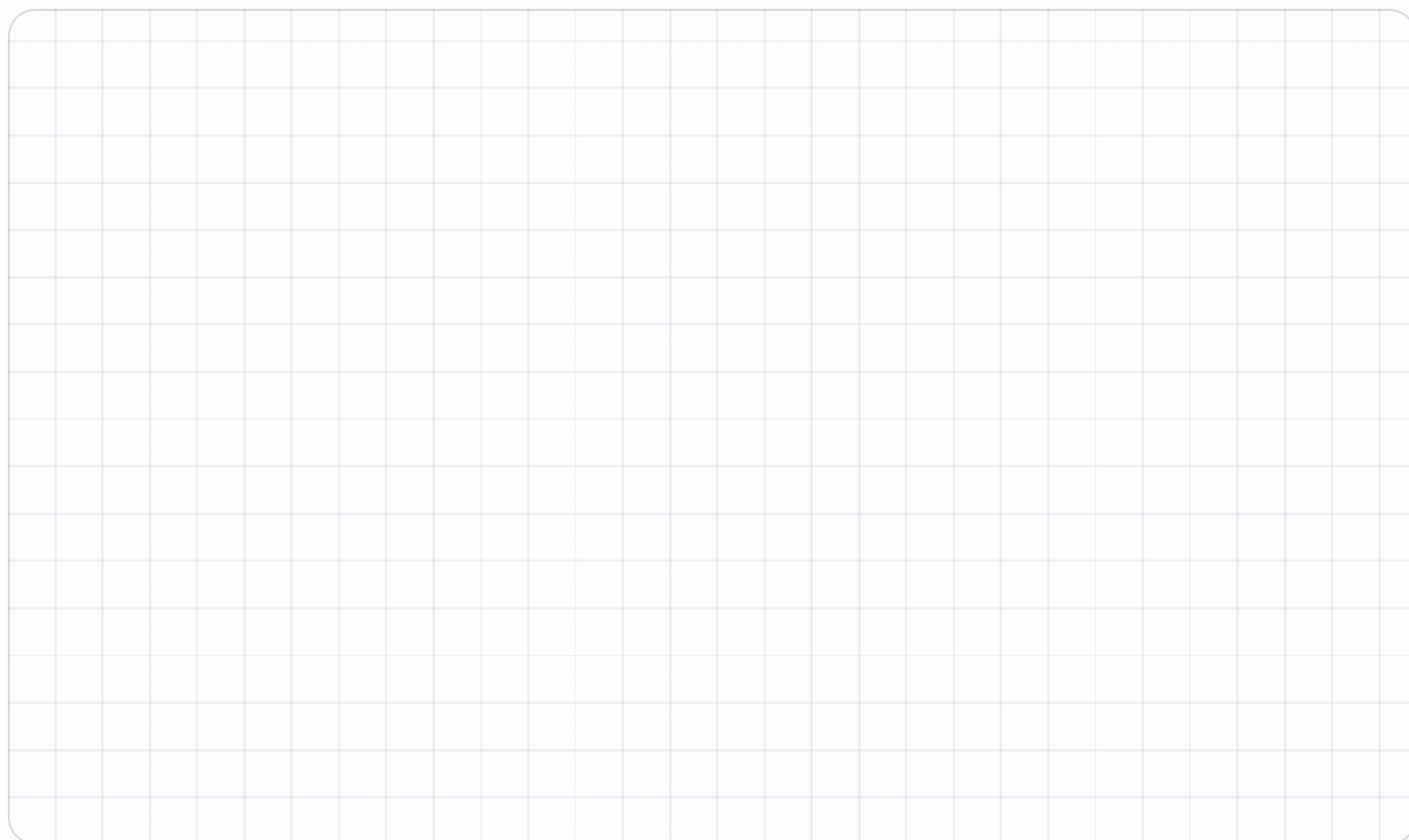
15.  $\frac{1}{4} + \frac{1}{4}$   
.....

16.  $\frac{2}{3} + \frac{1}{4}$   
.....

17.  $\frac{5}{6} + \frac{7}{12}$   
.....

18.  $\frac{7}{8}$  of 176  
.....

19.  $\frac{2}{3}$  of  $\frac{3}{5}$  — type as a/b  
.....

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Week 6 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 6.1 • The Equivalence Proof

OUT OF 15

Warm-Up • 1-8 • 2 |  $\frac{1}{2}$  |  $\frac{3}{6}$  |  $\frac{1}{2}$  | 1 | 0 | 8 | 6

Core • 9-15 •  $\frac{8}{12}$  |  $\frac{2}{3}$  |  $\frac{15}{18}$  |  $\frac{5}{6}$  | 4 | 18 |  $\frac{1}{2}$

Challenge • 16-26 •  $\frac{2}{3}$  |  $\frac{2}{3}$  | 1 |  $\frac{1}{2}$  |  $\frac{1}{3}$  |  $\frac{1}{2}$  |  $\frac{5}{6}$  |  $\frac{5}{12}$  |  $\frac{7}{6}$  |  $\frac{17}{12}$  | 95

## 6.2 • Number Line Defence

OUT OF 6

Warm-Up • 1-4 • 4 | 9 | 6 | 7

Core • 5-6 • 33 |  $\frac{1}{3}$

Challenge • 7-26 • 0.5 | 0.25 | 0.75 | 0.2 | 0.4 | 0.1 | 0.375 | 0.625 | 0.7 |  $\frac{2}{3}$  |  
0.6 | 0.875 |  $\frac{1}{3}$  |  $\frac{3}{4}$  |  $\frac{2}{5}$  |  $\frac{9}{10}$  |  $\frac{7}{12}$  |  $\frac{7}{9}$  |  $\frac{41}{24}$  | 132

## 6.3 • Mission Review

OUT OF 18

Warm-Up • 1-10 •  $\frac{3}{4}$  | 7 |  $\frac{2}{3}$  |  $\frac{1}{2}$  |  $\frac{1}{4}$  |  $\frac{1}{8}$  |  $\frac{1}{2}$  | 1 | 20 | 10

Core • 11-18 • 24 | 24 | 25 |  $\frac{1}{6}$  |  $\frac{1}{12}$  |  $\frac{1}{3}$  |  $\frac{3}{8}$  |  $\frac{1}{10}$

Challenge • 19-26 •  $\frac{1}{2}$  | 0.5 |  $\frac{11}{12}$  |  $\frac{5}{12}$  | 1 | 35 |  $\frac{1}{2}$  |  $\frac{1}{2}$

## Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 •  $\frac{1}{2}$  |  $\frac{2}{3}$  | 4 | yes | 0.2 |  $\frac{2}{3}$  |  $\frac{3}{4}$  | 6 |  $\frac{2}{3}$  | 8 |  $\frac{1}{2}$  | 8

Core • 13-22 •  $\frac{5}{6}$  |  $\frac{3}{4}$  | 20 |  $\frac{3}{4}$  | 0.375 |  $\frac{11}{12}$  |  $\frac{3}{8}$  | 40 |  $\frac{7}{10}$  | 28

Challenge • 23-26 •  $\frac{17}{12}$  | 50 |  $\frac{29}{24}$  | 84

## Fri • Mission 04 Test

OUT OF 14

Warm-Up • 1-4 •  $\frac{1}{2}$  |  $\frac{1}{2}$  | 10 |  $\frac{1}{2}$

Core • 5-14 • 5 |  $\frac{2}{3}$  |  $\frac{3}{4}$  |  $\frac{5}{12}$  |  $\frac{5}{8}$  | 12 | 0.75 | 12 |  $\frac{11}{12}$  | 33

Challenge • 15-19 •  $\frac{17}{12}$  | 24 |  $\frac{17}{12}$  | 154 |  $\frac{2}{5}$